Q1. Principles and Techniques of Symmetric Cryptography [10 Marks]

Symmetric cryptography is an encryption technique in which the sender and receiver use the same secret key for encryption and decryption.

Main Components

1. Plaintext: The original readable message or data.
2. Encryption Algorithm: Converts plaintext into ciphertext using a secret key.
3. Secret Key: A confidential key shared between sender and receiver.
4. Ciphertext: The unreadable, encrypted form of the original message.
5. Decryption Algorithm: Converts ciphertext back into plaintext using the secret key.

The encryption and decryption formulas are:

\[C=E(K,M) \]

\[M=D(K,C) \]

Where \(M\) = plaintext, \(C\) = ciphertext, and \(K\) = secret key.

Kerckhoffs's Principle

Kerckhoffs's principle states that a cryptographic system should remain secure even if everything about the system is publicly known except the secret key. Security should depend on the secrecy of the key, not on hiding the algorithm.

Labelled Block Diagram

Sender: Plaintext (M)

Encryption Algorithm

Secret Key (K)

Ciphertext (C)

Decryption Algorithm

Same Secret Key (K)

Receiver: Original Plaintext (M)

Advantages: Fast, efficient, and suitable for large amounts of data.

Limitation: The secret key must be distributed securely.

Q2. Stream Ciphers, Block Ciphers, Confusion, Diffusion and Modes [10 Marks]

1. Stream Cipher vs Block Cipher

Basis	Stream Cipher	Block Cipher
Operation	Encrypts data bit by bit or byte by byte	Encrypts fixed-size blocks

Speed	Generally fast	Fast for bulk data
Padding	Usually not required	Depends on the mode
Examples	ChaCha20, RC4	AES, DES
Applications	Real-time communication	File and database encryption
Main risk	Keystream or nonce reuse	Incorrect mode or IV/nonce use

RC4 and DES are now considered insecure for new systems.

2. Confusion and Diffusion

Confusion: It makes the relationship between the secret key and ciphertext complex. Substitution boxes (S-boxes) provide confusion.

Diffusion: It spreads the effect of one plaintext bit across many ciphertext bits, hiding statistical patterns. Permutations and mixing operations provide diffusion.

Both concepts were explained by Claude Shannon.

3. Modes of Operation

Mode	Working	Advantages	Disadvantages
ECB	Encrypts each block independently	Simple and parallel	Reveals repeated patterns
CBC	XORs plaintext with the previous ciphertext block before encryption	Hides repeated block patterns when used correctly	Requires proper IV and padding; encryption is sequential
CFB	Uses previous ciphertext to generate a stream-like output	Can process smaller units	Sequential and requires correct IV handling
OFB	Generates a keystream using repeated encryption	Transmission errors do not propagate through later plaintext	IV reuse repeats the keystream
CTR	Encrypts counter values and XORs the output with plaintext	Fast, parallel and no padding required	Counter/nonce must not repeat under the same key

Note: These modes do not independently provide authentication. Modern systems commonly use authenticated encryption such as AES-GCM.

Q3. Historical Development of Symmetric Encryption [10 Marks]

The development of symmetric encryption progressed from simple manual methods to advanced computer algorithms.

1. Caesar Cipher: An ancient cipher that shifts letters by a fixed number of positions. It was simple but easy to break.

2. Enigma Machine: Used by Germany during World War II. It employed rotating components to produce changing substitutions. Allied cryptanalysts successfully broke many Enigma communications.

3. Claude Shannon: In 1949, Shannon published Communication Theory of Secrecy Systems. He explained important concepts such as perfect secrecy, confusion and diffusion.

4. Feistel Cipher: Horst Feistel helped develop the Feistel structure, which divides a block into two halves and processes them through multiple rounds.

5. DES (1977): The Data Encryption Standard was adopted in 1977. It uses 64-bit blocks and an effective 56-bit key. Its short key later became vulnerable to brute-force attacks.

6. Triple DES (3DES): DES was applied three times to improve security. It was more secure than ordinary DES but slower and is now considered legacy technology.

7. IDEA: Introduced in 1991, IDEA uses a 128-bit key and a 64-bit block.

8. Blowfish: Introduced in 1993, it supports variable-length keys and 64-bit blocks.

9. RC4: A stream cipher widely used historically, but weaknesses led to its rejection for modern security applications.

10. AES (2001): NIST selected Rijndael in an open competition. AES uses 128-bit blocks and supports 128-, 192- and 256-bit keys.

Conclusion: Modern symmetric encryption uses stronger algorithms, larger keys and publicly evaluated designs. AES is preferred over DES and 3DES for new applications.

Q4. Classical Substitution Ciphers [10 Marks]

(a) Caesar Cipher

Each letter is shifted by a fixed number of positions.

Example: With a shift of 3:

- Plaintext: HELLO
- Ciphertext: KHOOR

Encryption formula:

$$ C=(P+K)\bmod 26 $$

Strengths: Simple and easy to implement.

Weaknesses: Very small key space; easily broken by brute force or frequency analysis.

(b) Monoalphabetic Substitution Cipher

Each plaintext letter is replaced by a fixed corresponding ciphertext letter. For example, A may always become Q and B may always become W.

Strengths: Has many possible substitution keys.

Weaknesses: Letter frequencies and word patterns remain similar, making frequency analysis effective.

(c) Playfair Cipher

Playfair encrypts pairs of letters using a 5 × 5 matrix constructed from a keyword. Usually, I and J share one position.

Rules:

1. If both letters are in the same row, take the letters to their right.
2. If both are in the same column, take the letters below them.
3. Otherwise, use the rectangle rule and take the other corners in the same rows.
4. Decryption reverses the row and column directions.

Strengths: Hides ordinary single-letter frequency patterns better than simple substitution.

Weaknesses: Digraph patterns remain and can be analysed.

(d) Vigenère Polyalphabetic Cipher

Vigenère uses a repeating keyword to apply different Caesar shifts to different letters.

\[C_i=(P_i+K_i)\bmod 26 \]

For example, the keyword determines the shift applied to each plaintext letter.

Strengths: Hides simple single-letter frequency patterns.

Weaknesses: Repeating keys can be identified using methods such as Kasiski examination.

Why Is Frequency Analysis Effective?

Caesar and monoalphabetic ciphers preserve letter-frequency patterns. Playfair encrypts pairs of letters, making ordinary frequency analysis harder. Vigenère uses multiple substitutions, but a repeating keyword introduces patterns that can eventually be detected.

Q5. Transposition, Product Ciphers and One-Time Pad [8 Marks]

1. Rail Fence Cipher

The rail fence cipher writes plaintext diagonally across several rows and then reads each row sequentially.

For example, using two rails:

Plaintext: HELLOWORLD

The letters are placed alternately on two rows, and each row is read to produce ciphertext.

Advantages: Simple and easy to understand.

Disadvantages: Preserves letter frequencies and is easy to break.

2. Columnar Transposition Cipher

In this method, plaintext is written row by row into a table. The columns are then read according to the alphabetical order of a keyword.

Advantages: Rearranges letter positions and hides the original word order.

Disadvantages: Preserves letter frequencies and can be attacked using statistical analysis.

3. Product Ciphers

A product cipher combines substitution and transposition operations.

- Substitution replaces symbols with other symbols.
- Transposition rearranges the positions of symbols.

Combining these operations repeatedly provides confusion and diffusion. Modern block ciphers such as AES use more advanced mathematical transformations based on similar principles.

4. One-Time Pad

A one-time pad is a symmetric encryption method that combines plaintext with a truly random secret key, commonly using XOR.

It is theoretically unbreakable when:

- The key is truly random.
- The key is as long as the message.
- The key remains secret.
- The key is used only once.

Why is it impractical? Generating, distributing and securely storing a message-length key is difficult. Reusing the key destroys its perfect-secrecy guarantee.

Q6. Data Encryption Standard (DES) Structure [12 Marks]

DES is a symmetric block cipher adopted in 1977. It encrypts 64-bit blocks using an effective key length of 56 bits.

1. Feistel Structure

DES divides the 64-bit input block into two 32-bit halves:

- L_0: Left half
- R_0: Right half

It performs 16 rounds of processing.

The round equations are:

$$ L_i=R_{i-1} $$

$$ R_i=L_{i-1}\oplus F(R_{i-1},K_i) $$

Here, K_i is the round key and F is the round function.

2. Main Components of DES

Initial Permutation (IP): Rearranges the 64 input bits according to a fixed table.

Sixteen Rounds: Each round applies the Feistel transformation using a different round key.

Final Permutation (FP): Applies the inverse of the initial permutation to produce ciphertext.

3. DES Round Function

The round function contains four main steps:

1. Expansion: Expands the 32-bit right half to 48 bits.
2. Key Mixing: XORs the expanded result with a 48-bit round key.
3. S-box Substitution: Eight S-boxes convert the 48-bit result into 32 bits, providing confusion.
4. P-box Permutation: Rearranges the 32 bits, helping diffusion.

4. Key Schedule

The original key is represented by 64 bits, including 8 parity bits. The remaining 56 bits are processed to generate sixteen 48-bit round keys.

The key schedule performs permutations, divides the key into two halves, rotates the halves and selects bits to form each round key.

Labelled Diagram of One DES Round

Right half R_{i-1} — 32 bits

Expansion E: 32 → 48 bits

XOR with round key K_i — 48 bits

S-boxes: 48 → 32 bits

P-box permutation — 32 bits

XOR with L_{i-1} to produce R_i

The next left half is $L_i=R_{i-1}$.

Decryption: DES uses the same Feistel structure but applies the round keys in reverse order.

Limitation: DES is no longer secure because its 56-bit key is too short against modern brute-force attacks.

Q7. Strengths and Weaknesses of DES, Double DES and Triple DES [10 Marks]

1. Strengths of DES

- Uses a well-defined 16-round Feistel structure.
- Produces a strong avalanche effect.
- Its S-boxes provide nonlinear transformations.
- Was widely studied and implemented.

Avalanche effect: A small change in the plaintext or key should change approximately half of the ciphertext bits.

2. Weaknesses of DES

Short key: The effective key length is only 56 bits, giving 2^{56} possible keys.

Brute-force attack: An attacker can try keys until the correct key is found.

Differential cryptanalysis: Studies how differences in plaintext affect differences in ciphertext.

Linear cryptanalysis: Uses approximate linear relationships between plaintext, ciphertext and key bits.

Small block size: Its 64-bit blocks are a limitation when encrypting large volumes of data.

3. Double DES

Double DES encrypts data twice using two keys:

$$ C=E(K_2,E(K_1,P)) $$

It was intended to increase security beyond ordinary DES.

Meet-in-the-Middle Attack

An attacker calculates $E(K_1,P)$ for possible first keys and $D(K_2,C)$ for possible second keys. Matching intermediate values reveal candidate key pairs.

Therefore, the attack requires roughly 2^{56}-scale work with substantial storage rather than the expected 2^{112} exhaustive search.

4. Triple DES (3DES)

Triple DES applies DES three times, commonly in Encrypt–Decrypt–Encrypt order:

$$ C=E(K_3,D(K_2,E(K_1,P))) $$

- Two-key 3DES: Uses $K_1=K_3$.
- Three-key 3DES: Uses three independent keys.

Triple DES offers greater security than DES, but it is slower and retains the 64-bit block-size limitation. It is now considered legacy technology.

Conclusion: AES is preferred for modern applications because it supports larger keys, has a 128-bit block size and performs efficiently.

Q8. Advanced Encryption Standard (AES) [10 Marks]

AES is a symmetric block cipher standardised by NIST in 2001. It is based on the Rijndael algorithm.

1. AES Block Size and Key Lengths

AES always uses a 128-bit block size.

Key Length	Number of Rounds
128 bits	10
192 bits	12
256 bits	14

2. Four AES Round Transformations

1. SubBytes: Each byte is replaced using a nonlinear S-box. This provides confusion.

2. ShiftRows: Rows of the internal 4 × 4 byte state are shifted cyclically by different offsets. This spreads byte positions.

3. MixColumns: Each column is mathematically transformed using finite-field arithmetic. This provides diffusion.

4. AddRoundKey: The state is XORed with a round key derived from the original secret key.

3. AES Encryption Process

1. The plaintext is arranged into a 4 × 4 byte state.
2. An initial AddRoundKey operation is performed.
3. The main rounds apply SubBytes, ShiftRows, MixColumns and AddRoundKey.
4. The final round performs SubBytes, ShiftRows and AddRoundKey, omitting MixColumns.
5. The resulting state becomes the ciphertext.

4. Key Expansion

The AES key expansion process generates round keys from the original secret key. It uses word rotations, S-box substitutions and round constants. Each round receives the required round-key material.

5. AES Decryption

AES decryption reverses the encryption transformations using:

- InvShiftRows
- InvSubBytes
- AddRoundKey
- InvMixColumns

These inverse operations are applied in the correct reverse order, using the expanded round keys.

6. Comparison Between DES and AES

Feature	**DES**	**AES**
Structure	Feistel network	Substitution–permutation network
Block size	64 bits	128 bits
Key length	56 effective bits	128, 192 or 256 bits
Number of rounds	16	10, 12 or 14
Security	Insecure for modern use	Strong when properly implemented

Performance	Legacy; 3DES is slow	Efficient in modern hardware and software
Current use	Obsolete	Recommended for new systems

Conclusion: AES provides stronger modern security than DES. It supports larger key sizes, a larger block size and efficient implementation. For practical applications, AES should be used with a secure authenticated-encryption mode such as AES-GCM and proper key management.